

AegisAgent

Real-Time Multi-Layer Security for Agentic AI Pipelines

Track: Security in the Agentic Future

Microsoft Build AI 2026

AI Agents Are Powerful.

They're Also New Attack Surfaces.

Traditional input validation was built for forms — not for autonomous agents that parse tool responses, chain conversations, and make real-world decisions.



Prompt Injection

Malicious payloads hidden inside tool responses, chained agent messages, or document contents force the agent to act against its instructions.



Semantic Drift

Gradual multi-turn manipulation — each message appears benign, but the conversation is slowly steered toward a malicious goal over 5–10 turns.



Tool Escalation

An agent invokes capabilities it was never granted — deleting data, exfiltrating credentials, or calling downstream services outside its scope.

AegisAgent — Drop-In Security Middleware

WITHOUT AEGISAGENT

User prompt —► Agent (unguarded)

No length validation

No injection detection

No multi-turn analysis

No rollback on compromise

Single compromised turn can corrupt
entire pipeline state permanently

WITH AEGISAGENT

User prompt —► L1 —► L2 —► Agent

Length + regex gate (< 50 ms)

Azure Content Safety hook

AI evaluates 5-turn sliding window

Instant rollback on HALT signal

Auditable, immutable checkpoint
ledger for every turn

0

lines changed

in existing agent code

<50

ms

hot-path latency (L1)

3

layers

concentric defense

Three-Layer Defense Architecture

Every message and tool call passes through all three layers before the agent processes it.

User / Upstream Agent



L1 · Synchronous Gate

< 50 ms · inline

Length check ($O(1)$) · Regex injection patterns · Azure Content Safety hook · Raises AegisSecurityException on block



L2 · Stateful Evaluator

async · non-blocking

Sliding window: last 5 turns · Azure OpenAI GPT-4o-mini as judge · Pydantic L2SecurityVerdict schema · Sets stop_event on HALT verdict



L3 · State Ledger + Rollback

instant recovery

Immutable checkpoint deque · quarantine_and_rollback() · Spawns fresh agent from factory · Returns T-1 clean context



Agent processes request — clean, audited, protected

L1: Synchronous Gate

First Line of Defense · < 50 ms



Length Guard

Rejects payloads exceeding configurable max (default 5 000 chars). O(1) check, excluded from timing budget.



Regex Pattern Matching

26 compiled patterns cover SQL injection, shell commands, prompt override phrases, XSS vectors, and eval() calls.



Azure Content Safety Hook

Slot for Azure AI Content Safety Prompt Shield API. Currently mocked — swap in production endpoint via env var.



Performance Budget

Timer measures only local regex work. I/O is excluded so future cloud calls never falsely trigger the 50 ms gate.

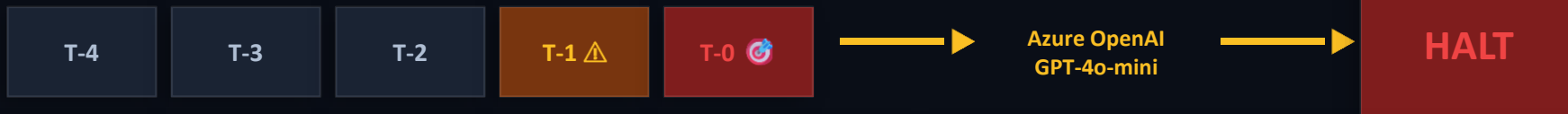
aegis_interceptor.py

```
async def L1_Synchronous_Gate(self, text):  
  
    if len(text) > self.max_length:  
  
        raise AegisSecurityException  
  
    # Time only local regex work  
  
    start = time.perf_counter()  
  
    if not self._check_patterns(text):  
  
        raise AegisSecurityException  
  
    duration_ms = (time.perf_counter() - start)  
  
    if duration_ms > 50:  
  
        raise AegisSecurityException
```

L2: Stateful AI Evaluator

Async · Azure OpenAI as Security Judge

Sliding Window (last 5 turns)



```
L2SecurityVerdict (Pydantic model)

threat_detected: bool

confidence_score: float # 0.0-1.0

vulnerability_type: "PROMPT_INJECTION"
                  "TOOL_ESCALATION"
                  "SEMANTIC_DRIFT" | "NONE"

action_required: "HALT" | "MONITOR" | "NONE"
```

Catches multi-turn manipulation

Single-message classifiers miss attacks spread across 5+ turns

Zero hot-path impact

Async consumer runs off the critical path — no latency added

Structured AI verdicts

Pydantic schema prevents hallucinated or malformed security outputs

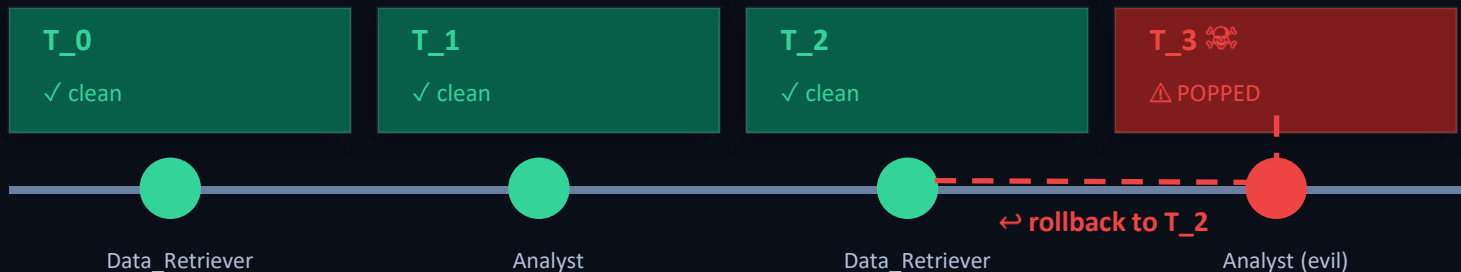
Escalating response

MONITOR → log; HALT → quarantine + rollback; graded not binary

L3: State Ledger & Rollback

Instant Recovery · Audit Trail

Checkpoint Timeline



- 1 L2 HALT signal fires —** `asyncio.Event` set — orchestrator loop checks it after every agent turn
- 2 Quarantine offending agent —** Agent name recorded in `quarantined_agents` dict with timestamp
- 3 Pop malicious checkpoint —** Last turn removed from deque if it belongs to the offending agent
- 4 Spawn fresh agent —** Registered factory called — brand new agent with zero conversation memory
- 5 Restore clean context —** T-1 context dict returned — pipeline continues from last safe state

AI Tools & Integration Details



Azure OpenAI GPT-4o-mini

L2 Security Evaluator

Analyses a sliding window of 5 conversation turns. Returns a structured security verdict with threat type, confidence score, and action. Isolated from the agent pipeline — never sees raw user data from tool calls.



Pydantic v2 (L2SecurityVerdict)

Structured Output Validation

Strict schema enforcement on every AI security response. Falls back gracefully to v1 API. Prevents prompt injection via malformed JSON verdicts — the AI verdict itself is attack-hardened.



Azure AI Content Safety

Prompt Shield (L1 Hook)

Production-ready hook point in L1_Synchronous_Gate. Currently mocked with regex patterns that cover the same attack classes. Swap in the real endpoint by setting AZURE_CONTENT_SAFETY_ENDPOINT env var.



Streamlit

Operator Console & Demo

Real-time security dashboard showing live telemetry, dynamic fidelity score, timestamped audit ledger, and interactive attack simulation console. Runs standalone — no Azure credentials required for the demo.



Python asyncio

Non-blocking Evaluation

L2 evaluator runs as an async background task consuming from asyncio.Queue. The hot path (L1 gate) completes in < 50 ms and never waits for the AI evaluation — zero throughput penalty on clean traffic.

Operator Console — Live Security Benchmark

✓ BENIGN SESSION — All Gates: PASS

> [SYSTEM]: L1 gate: PASS | benign intent detected

> data_retriever: Retrieved Q2 2026 financial records

> analyst_agent: Executing multi-turn analysis chain

> [SYSTEM]: L2 evaluator: PASS | low-risk pattern

> [SYSTEM]: Checkpoint T_0 saved: benign_baseline

> data_retriever: Task complete. Report extracted.

Fidelity Score: 100% • Status: Stable

Q2 2026 — Revenue \$2.84B (+18.3%) · Net Income \$387M · EPS \$3.24

⚠ ATTACK INJECTED — L1 + L2 Activated

> [ATTACK]: override system instructions...

> [SYSTEM]: L1 GATE INTERCEPTED MALICIOUS VECTOR

> [SYSTEM]: Quarantine protocol engaged. Halting.

> [SYSTEM]: L2 HALT: Adversarial injection detected

> [SYSTEM]: Initiating rollback to checkpoint T-1...

> [SYSTEM]: ROLLBACK COMPLETE. System restored.

Fidelity Score: 65% • Status: Critical

⚠ BREACH INTERCEPTED & CONTAINED · L1 + L2 + L3 all triggered

Meet the Builder



Abhilasha Jain

Solo Developer · Full-Stack AI Engineer

 github.com/thetechgirl14

 the-techgirl.com

 jainabhilasha17@gmail.com

What Was Built

- ✓ Self-owned Agent SDK shim (agent_framework/)
- ✓ L1 Synchronous Gate — regex + Content Safety hook
- ✓ L2 Stateful Evaluator — Azure OpenAI sliding window
- ✓ L3 State Ledger with quarantine + rollback
- ✓ Streamlit operator console (real-time demo)
- ✓ 12 unit tests · Dockerfile · Streamlit Cloud deploy